

## Appendix – D3 & D4 Consistency and Final Review

### D3 – Consistency Check

This section documents the consistency verification conducted across the 12 selected use cases. Each use case was checked for the presence of its realization artifact, correctness of extracted methods against the final Design Class Diagram (A5), and compliance with the defined layered architecture.

| UC ID | UC Name                         | CRC | Comm | Seq | Methods in DCD | Layering OK |
|-------|---------------------------------|-----|------|-----|----------------|-------------|
| UC-01 | Book Appointment                | -   | -    | ✓   | ✓              | ✓           |
| UC-02 | Register Patient Information    | ✓   | -    | -   | ✓              | ✓           |
| UC-03 | Manage Doctor Schedule          | -   | -    | ✓   | ✓              | ✓           |
| UC-04 | Confirm Appointment Slot        | -   | ✓    | -   | ✓              | ✓           |
| UC-05 | Complete Patient Treatments     | -   | -    | ✓   | ✓              | ✓           |
| UC-06 | Cancel Appointment              | ✓   | -    | -   | ✓              | ✓           |
| UC-07 | Make Payment                    | -   | -    | ✓   | ✓              | ✓           |
| UC-08 | Authorize Patient Discharge     | -   | ✓    | -   | ✓              | ✓           |
| UC-09 | Send Appointment Reminders      | ✓   | -    | -   | ✓              | ✓           |
| UC-11 | Generate Monthly Bill           | -   | ✓    | -   | ✓              | ✓           |
| UC-T2 | Update Slot Availability        | ✓   | -    | -   | ✓              | ✓           |
| UC-T3 | Synchronize Patient Information | -   | ✓    | -   | ✓              | ✓           |

No missing realization artifacts, method inconsistencies, or architectural layer violations were identified during the D3 consistency check.

## D4 – Final Diagram Review and Sign-Off

A final review pass was performed on all diagrams prior to freezing the technical report (A9). This review verified naming consistency, completeness of method definitions with respect to A5, and strict adherence to architectural layering constraints.

| Diagram              | Type          | Scope                    | Names OK | Methods OK | Layers OK | Fix Needed |
|----------------------|---------------|--------------------------|----------|------------|-----------|------------|
| Design Class Diagram | Class         | All UCs                  | ✓        | ✓          | ✓         | No         |
| UC-01                | Sequence      | Book Appointment         | ✓        | ✓          | ✓         | No         |
| UC-02                | CRC           | Register Patient         | ✓        | ✓          | ✓         | No         |
| UC-03                | Sequence      | Manage Schedule          | ✓        | ✓          | ✓         | No         |
| UC-04                | Communication | Confirm Slot             | ✓        | ✓          | ✓         | No         |
| UC-05                | Sequence      | Complete Treatment       | ✓        | ✓          | ✓         | No         |
| UC-06                | CRC           | Cancel Appointment       | ✓        | ✓          | ✓         | No         |
| UC-07                | Sequence      | Make Payment             | ✓        | ✓          | ✓         | No         |
| UC-08                | Communication | Authorize Discharge      | ✓        | ✓          | ✓         | No         |
| UC-09                | CRC           | Send Reminders           | ✓        | ✓          | ✓         | No         |
| UC-11                | Communication | Generate Bill            | ✓        | ✓          | ✓         | No         |
| UC-T2                | CRC           | Update Slot Availability | ✓        | ✓          | ✓         | No         |
| UC-T3                | Communication | Synchronize Patient      | ✓        | ✓          | ✓         | No         |

Based on this final review, all diagrams are internally consistent, complete, and compliant with the defined architectural and naming standards. The documentation is ready for final PDF submission.